

Week 2 module

① Fraud Transaction Detection (Linear search) Java program

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] transactions = new int[N];
```

```
        for (int i = 0; i < N; i++) {
            transactions[i] = sc.nextInt();
        }
        int suspicious = sc.nextInt();
        boolean found = false;
```

```
        for (int i = 0; i < N; i++) {
            if (transactions[i] == suspicious) {
                found = true;
                break;
            }
        }
```

```
        if (found) {
            System.out.println("Fraud Transaction found");
        }
        else {
            S.o.p ("Transaction not found");
        }
    }
}
```

Sorting5.java / ...

```
1
2 import java.util.Scanner;
3
4 public class Sorting5{
    Run main | Debug main | Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int N = sc.nextInt();
9         int[] arr = new int[N];
10
11         for (int i = 0; i < N; i++) {
12             arr[i] = sc.nextInt();
        }
```

PROBLEMS 19 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ac Sorting5.java } ; if ($?) { java Sorting5 }
```

```
5
```

```
1 2 5 8 9
```

```
1 2 5 8 9
```

```
PS C:\Users\Admin\OneDrive\Desktop\New java> |
```


② server log search (Binary search)

```

int N = sc.nextInt();
int[] logs = new int[N];
for (int i = 0; i < N; i++) {
    logs[i] = sc.nextInt();
}
int target = sc.nextInt();
int low = 0, high = N - 1;
boolean found = false;
while (low <= high) {
    int mid = (low + high) / 2;
    if (logs[mid] == target) {
        found = true;
        break;
    }
    else if (logs[mid] < target) {
        low = mid + 1;
    }
    else {
        high = mid - 1;
    }
}
if (found) {
    System.out.println("Log found");
}
else {
    s.o.p("Log not found");
}
sc.close();
}
}

```

Sorting4.java > ...

```
4  public class Sorting4{
5      public static void main(String[] args) {
17          int key = arr[1];
18          int j = i - 1;
19
20          while (j >= 0 && arr[j] > key) {
21              arr[j + 1] = arr[j];
22              j--;
23          }
24
25          arr[j + 1] = key;
26      }
27
28      for (int i = 0; i < N; i++) {
29          System.out.print(arr[i] + " ");
30      }
31
32      sc.close();
33  }
34  }
35
36
```

PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ac Sorting4.java } ; if ($?) { java Sorting4 }
```

5

```
240 245 569 890 908 789
```

```
240 245 569 890 908
```

```
PS C:\Users\Admin\OneDrive\Desktop\New java>
```


④ sort website Response Time (insertion sort)

```
int N = sc.nextInt()
```

```
public class Main {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int N = sc.nextInt();
```

```
        int[] arr = new int[N];
```

```
        for (int i = 0; i < N; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        // Insertion sort
```

```
        for (int i = 1; i < N; i++) {
```

```
            int key = arr[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && arr[j] > key) {
```

```
                arr[j+1] = arr[j];
```

```
                j--;
```

```
            }
```

```
            arr[j+1] = key;
```

```
        }
```

```
        for (int i = 0; i < N; i++) {
```

```
            System.out.print (arr[i] + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```
2 import java.util.Scanner;
3
4 public class Sorting3 {
    Run main | Debug main | Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int N = sc.nextInt();
9         int[] logs = new int[N];
10
11         for (int i = 0; i < N; i++) {
12             logs[i] = sc.nextInt();
13         }
14
15         int target = sc.nextInt();
16
17         int low = 0, high = N - 1;
18         boolean found = false;
19
20         while (low <= high) {
21             int mid = (low + high) / 2;
22
23             if (logs[mid] == target) {
24                 found = true;
25                 break;
            }
        }
    }
}
```

PROBLEMS 17 OUTPUT DEBUG CONSOLE TERMINAL PORTS

1001 1005 1010 1015 1020 1030 1040

1015

Log Found

PS C:\Users\Admin\OneDrive\Desktop\New java>

⑤ prioritize Bug Reports (selection sort)

```
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int N = sc.nextInt();
        int[] arr = new int[N];
        for (int i = 0; i < N; i++) {
            arr[i] = sc.nextInt();
        }
        // selection sort
        for (int i = 0; i < N - 1; i++) {
            int minIndex = i;
            for (int j = i + 1; j < N; j++) {
                if (arr[j] < arr[minIndex]) {
                    minIndex = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
        for (int i = 0; i < N; i++) {
            System.out.print (arr[i] + " ");
        }
        sc.close();
    }
}
```

```

1
2     import java.util.Scanner;
3
4 public class Sorting2{
5
6
7     Run main | Debug main | Run | Debug
8     public static void main(String[] args) {
9
10         Scanner sc = new Scanner(System.in);
11
12
13         int N = sc.nextInt();
14         int[] salaries = new int[N];
15
16
17         for (int i = 0; i < N; i++) {
18             salaries[i] = sc.nextInt();
19         }
20
21         // Bubble Sort
22         for (int i = 0; i < N - 1; i++) {
23             for (int j = 0; j < N - i - 1; j++) {
24                 if (salaries[j] > salaries[j + 1]) {
25                     int temp = salaries[j];
26                     salaries[j] = salaries[j + 1];
27                     salaries[j + 1] = temp;
28                 }
29             }
30         }
31     }
32 }

```

PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
c Sorting2.java } ; if ($?) { java Sorting2 }
```

```
5000 34600 55000 56789 70999 90870
```

```
4600 45000 55000 56789 70999
```

```
S C:\Users\Admin\OneDrive\Desktop\New java>
```


Sorting1.java > ...

```
3 public class Sorting1{
5 public static void main(String[] args) {
8
9     int N = sc.nextInt();
10    int[] transactions = new int[N];
11
12    for (int i = 0; i < N; i++) {
13        transactions[i] = sc.nextInt();
14    }
15
16    int suspicious = sc.nextInt();
17    boolean found = false;
18
19    for (int i = 0; i < N; i++) {
20        if (transactions[i] == suspicious) {
```

PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
c: Sorting1.java } ; if ($?) { java Sorting1 }
```

209 7890 6709 1404 1234

234

Fraud Transaction Found